DOD-P-63245A (AR)
28 September 1981
SUPERSEDING
DOD-P-63245 (AR)
10 January 1979

MILITARY SPECIFICATION

PROJECTILE, H.E., 155MM, XM795
METAL PARTS ASSEMBLY

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the manufacturing and assembling of metal parts for Projectile, H.E., 155MM, XM795.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	-Cleaning Methods and Pre-treatment of Ferrous Surfaces for Original Coatings

MILITARY

MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-C-16173	-Corrosion Preventive Compound, Solvent Cutback, Cold-Application
MIL-M-6867	-Magnetic Inspection Units
MIL-I-6868	-Inspection Process, Magnetic Particle
MIL-S-50783	-Steel Alloy; Special Purpose for Ammunition Components (HF-1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-1169	-Packaging, Packing and Marking of Inert Ammunition
MIL-STD-109	-Quality Assurance Terms & Definitions
MIL-STD-410	-Nondestructive Testing Personnel Qualification & Certification
MIL-STD-1459	-Macrograph Standards for Steel Bars, Billets, and Blooms for Ammunition Components

DRAWINGS (see 6.10)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9326055	-Projectile, 155MM, H.E., XM795 Metal Parts Assembly
9280308	-Ultrasonic Inspection Standard, Longitudinal and Radial Cracks: XM795 Body
9280309	-Ultrasonic Inspection Standard Circumferential and Piping Defects 155MM, XM795
9280336	-Envelope Drawing, Ultrasonic Inspection Equipment and Procedures for Body, Machined, 155MM, XM795
9329574	-Packing and Marking for Pallet Wood Ammunition for Projectile, 155MM, HE, XM795
9329766	-Body Configuration for Ultrasonic Inspection

PUBLICATIONS

ARRADCOM

Operating Instructions for use of Detroit Testing Machine for Band Seating Inspection

2.2 Other publications. The following documents form a part of this specification to the extent specified herein unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

ANSI-B46.1 -Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, New York 10018).

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ASTM E-8 -Tension Testing of Metallic Materials
ASTM B-117 -Salt Spray (Fog) Testing

(Application for copies should be sent to the American Society for Testing and Materials, 916 Race Street, Philadelphia, PA., 19103).

3. REQUIREMENTS

3.1 Material. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Projectile metal parts assembly. The projectile assembly and components shall comply with all requirements specified on Drawing (dwg.) 9326055 and with all requirements specified in applicable specifications.

3.3 Steel quality. The item contractor shall establish procedures and provide a forge quality steel that meets all requirements of MIL-S-50783 (MU). The quality of the steel as indicated by the results of the macroetch tests shall be equal to or better than macrograph standards A5, B3 and C8 of MIL-STD-1459 with defects D1, D3, D4, D5, D6, D7 and D8 unacceptable. The manufacturer will provide a certification to cite that all requirements of MIL-S-50783 (MU) have been met. Billet separation by flame cutting is not permitted.

3.4 Heat treatment. Prior to finish machining, projectile body forgings shall be subjected to a quench-temper heat treatment to attain the required mechanical properties. Heat treatment shall be uniform throughout each lot of projectiles and shall be so performed and controlled as to produce the mechanical properties specified on the applicable drawing. The heat treatment shall consist of heating above the critical temperature, not to exceed 840 degrees Celsius (° C), followed by an oil quench. After quenching, the body forging shall be tempered without delay at a temperature above 400 degrees Celsius (° C), suitable to produce required mechanical properties. After removal from the tempering furnace, the body forgings shall be cooled in air (forced air permitted) to a temperature of not more than 370 degrees Celsius (° C), after which cooling may be accelerated by water spray. Bodies may be reheat treated a maximum of two times. Contractor shall submit his heat treatment and reheat treatment control procedures to the Government technical agency prior to implementation.

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3.5 Hydrostatic test. The projectiles shall withstand an internal hydrostatic pressure in accordance with requirements specified on the applicable drawing without evidence of leakage, cracks, rupturers or permanent distortion.

3.6 Proving ground. The projectiles, when tested as specified, shall show no loss or break-up of metal components.

3.7 Mechanical properties

3.7.1 Body. Mechanical properties of the machined body as required by applicable drawing shall apply. A slow bend precracked charpy test will be performed for information purposes.

3.7.2 Rotating band. After application and subsequent removal, the rotating band shall have the following mechanical properties:

a. Tensile strength - 206.8 Megapascals (MPa) (30,000 psi) minimum (min)

b. Elongation - 25 percent min. in two centimeters of gage length

3.8 Rotating band

3.8.1 Band seat. Prior to banding, the band seat shall be clean and free from oil, grease, dirt, rust, or other foreign matter. Shot or sandblasting shall not be permitted for cleaning the band seat after knurling.

3.8.2 Band blank - (Gilding metal). The rotating band blank shall be in the form of a solid ring, annealed if necessary to comply with the applicable specification; except that when hot banding at not less than 427 degrees Celsius (^{o}C), no prior annealing is necessary.

3.8.3 Application. The bands may be applied by either hot or cold process but if applied hot, the bands shall not be heated above 870 degrees Celsius (^{o}C) to prepare them for application to the body. The bands shall be heated uniformly by a process which will prevent oxidation (loose scale) and applied in a manner that will hold the formation of scale between the band and the band seat to a minimum.

3.8.4 Band seating sidewall contact. The rotating band shall be applied after final heat treatment of the body, by carefully pressing the band blank into the band seat so as to

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make contact with the sidewalls for a minimum of eight-tenths of the band seat depth. The band shall be applied in such a manner as not to distort the body wall. In addition, there shall be no surface gap between the rotating band and the sidewalls of the band seat, after the rotating band has been machined to final dimension.

3.8.5 Band seating tightness. The diametrical clearance between the rotating band and the band seat shall not exceed 0.23 millimeter when non-destructive method of testing is used.

3.9 Metal defects

3.9.1 Body. The body, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects.

3.9.2 Body cavity. The body cavity shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, folds, pits and sharp edges.

3.9.3 Visual standards. Visual standards for evaluation of metal defects will be established by the contractor with guidance from Government Agency for each contract after the first month's production. However, establishment of visual standards will not waive the requirements of compliance with 3.9.1 and 3.9.2.

3.10 Ultrasonic inspection. The projectiles shall be free of any discontinuities equal to or in excess of the inspection criteria within their respective zones. The inspection criteria are notches and holes in standards (drawings 9280308 and 9280309) which simulate rejectable cracks and piping flaws. (That is, all projectiles shall have an echo signal that is less than the signal obtained from the standards).

3.11 Interior cavity finish

3.11.1 The interior cavity after painting shall be free from bare spots, blisters, grease, dirt and other foreign matter.

3.11.2 There shall be no pool of paint in the bottom of the cavity prior to palletizing.

3.12 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

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3.13 Tool control dimensions. Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the component dimensions prior to start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool.

3.14 Base cover weld. The weld shall be sound, and the welding operation shall be controlled for adequate fusion between projectile body and base cover.

3.15 Magnetic particle inspection. The projectile shall be free from cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects.

3.16 Workmanship. The requirements for workmanship are as shown on the applicable drawings and specifications and the following:

3.16.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and specifications. Cleaning and preparation are required by component details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.16.2 Painting. Painting shall comply with the requirements of the applicable drawings and specifications. All paint shall be dry to the touch before packing for shipment. Drying time prior to testing shall be in accordance with the applicable specification.

3.16.3 Surface finish. The requirements for surface finish are as detailed on the applicable drawing. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determinations.

3.16.4 Threads. Threads shall be full and undamaged for the entire minimum length or depth as specified on the applicable drawing, and shall be in accordance with the provisions of specification MIL-A-2550.

3.16.5 Burr. No part shall have a burr which might interfere with the assembly or function of the projectile or which might be injurious to personnel handling the item.

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3.16.6 Foreign matter. No part or assembly shall contain dirt, grease, rust, corrosion or other foreign matter.

3.16.7 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

3.16.8 Corrosion preventive compound compatibility. Corrosion preventive compound specified on the applicable drawing shall be compatible with the high explosive filler to be used. Should the use of material other than those indicated in 6.8 be desired, information relative to compatibility can be obtained from the responsible technical agency by request through the procuring activity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Projectile, 155MM, HE, XM795 Metal Parts Assy	9326055	45
Body Assembly	9326054	5
Blank Rotating Band	9326052	5
Body Machined	9326053	5

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Part Description	Drawings	Quantity
Body Machined (Prior to Cutting Threads and Band Seat	9326053	5
Obturator	10542907	5
Base Cover	9326051	5

All assemblies and components shall have been produced by the contractor or furnished by a supplier and shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized, by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.3.2 Inspections to be performed. The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to any or all the requirements of the applicable drawings and specifications and to the inspections of Table I. All inspections listed in Table I shall be classified Major unless otherwise specified in referenced paragraphs and specifications.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Projectile Assemblies and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile, 155MM, HE, XM795 Metal Parts Assembly (Dwg. 9326055)				
	Examination for Defects (Except Proving Ground Test)	25		3.2	4.4.2.8
	Proving Ground Test	20		3.6	4.5.9
	Body Assembly (Dwg. 9326054)				
	Examination for Defects	5		3.2	4.4.2.7
	Blank Rotating Band (Dwg. 9326052)				
	Examination for Defects	5		3.2	4.4.2.3
	Body, Machined (Dwg. 9326053)				
	Examination for Defects	5		3.2	4.4.2.6
	Body Configuration for Ultrasonic Inspection (Dwg. 9329766)				
	Examination for Defects	5		3.2	4.4.2.5

NOTES:

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Projectile Assemblies and Components				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Obturator (Dwg. 10542907)				
	Examination for Defects	5		3.2	4.4.2.2
	Base Cover (Dwg. 9326051)				
	Examination for Defects	5		3.2	4.4.2.1
	Maximum temperature of mult		4.4.3.8	3.2	4.4.2.4

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Heat treatment inspection lots of projectiles shall be defined as follows:

4.4.1.1 Standard procedure. A heat treatment lot of bodies shall consist of bodies from only one mill heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature and for the same time cycles, in one unchanged process.

4.4.1.2 Miscellaneous lot. Bodies manufactured from small heats of steel, or bodies that have become separated from previously delivered lots may be grouped into a miscellaneous heat treatment lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1. The maximum lot size shall be 2000 bodies.

4.4.1.3 Use. The heat treatment lot constituted per 4.4.1.1 or 4.4.1.2 shall be the basis for selecting samples for mechanical properties tests. Bodies in each heat treatment lot shall be identified to that particular lot until satisfactory mechanical properties tests and interior surface and internal radius examinations have been completed.

4.4.1.4 Projectile lot. Heat Treatment Lots that have been determined to have complied with yield strength and percent elongation requirements of the applicable drawing may be regrouped for assembly into projectile lots.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9326051
4.4.2.1	Base Cover				NEXT HIGHER ASSEMBLY 9326054
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined.				
MAJOR					
101	Thickness		0.40%	3.2	Gage
MINOR					
201	Diameter		0.65%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Obturator				10542907
					NEXT HIGHER ASSEMBLY
					9326055
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined.				
MAJOR					
101	True position of rear inside diameter with forward inside diameter		0.40%	3.2	Gage
102	Diameter outside, front		0.40%	3.2	Gage
103	Diameter outside, rear		0.40%	3.2	Gage
104	Diameter inside, front		0.40%	3.2	Gage
105	Depth of forward inside diameter		0.40%	3.2	Gage
106	Thickness of wall at rear		0.40%	3.2	Gage
107	Width		0.40%	3.2	Gage
108	Flashing irregularities		0.40%	3.2	Visual
109	Number of slots		0.40%	3.2	Visual
110	Min. breaking load	Note a)		3.2	Gage
MINOR					
201	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

a. Ten (10) obturators per five thousand (5,000).

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Blank, Rotating Band		9326052
			NEXT HIGHER ASSEMBLY
			9326054

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined.				
MAJOR					
101	Inside diameter		0.40%	3.2	Gage
102	Width		0.40%	3.2	Gage
103	Thickness		0.40%	3.2	Gage
MINOR	None Defined.				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Mult				9345132
					NEXT HIGHER ASSEMBLY
					9326053
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Temperature of mult, max.		4.4.3.8	3.2	Test

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Body Configuration for Ultrasonic Inspection				9329766
					NEXT HIGHER ASSEMBLY 9326053
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Ultrasonic Inspection		100%	3.10	4.5.3
2	Ultrasonic Test Stamp missing		100%	3.2	Visual
3	Metal Defects (exterior surface)		100%	3.9.1	Visual
4	Metal Defects (before nosing)		100%	3.9.2	4.5.2
MAJOR	None Defined.				
MINOR	None Defined.				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER 9326053
4.4.2.6	Body Machined				NEXT HIGHER ASSEMBLY 9326054
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal Defective (after surface treatment prior to painting)		100%	3.9	Visual
2	Hydrostatic Test		100%	3.5	4.5.1
3	Hydrostatic Test Stamp Missing		100%	3.2	Visual
MAJOR					
101	Diameter of rotating band seat		0.40%	3.2	Gage
102	Runout of rotating band seat		0.40%	3.2	Gage
103	Location of rotating band seat		0.40%	3.2	Gage
104	Width of rotating band seat		0.40%	3.2	Gage
105	Mechanical Properties of Body	(Note a		3.7.1	4.5.4
106	Thickness through the base		0.40%	3.2	Gage
107	Scoring or knurling missing from rotating band seat		0.40%	3.2	Visual
108	Depth of scoring		0.40%	3.2	Visual
109	Foreign matter on rotating band seat		0.40%	3.8.1	Visual
110	Runout of pitch diameter of thread with bourrelet and front face		0.40%	3.2	Gage
111	Runout of ogive with bourrelet (three locations)		0.40%	3.2	Gage
112	Runout of bourrelet relief with bourrelet		0.40%	3.2	Gage
113	Runout of boattail with bourrelet		0.40%	3.2	Gage

NOTES:
a. See Paragraph 4.4.3.1

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CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.6	Body Machined		9326053
			NEXT HIGHER ASSEMBLY 9326054

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
114	Diameter of bourrelet, min.		0.40%	3.2	Gage
115	Diameter of bourrelet relief		0.40%	3.2	Gage
116	Length from front face to first basic diameter		0.40%	3.2	Gage
117	Length from front face to second basic diameter		0.40%	3.2	Gage
118	Diameter at 186.34mm from forward end		0.40%	3.2	Gage
119	Diameter at 313mm from forward end		0.40%	3.2	Gage
120	Length from rear face to forward basic diameter on boattail		0.40%	3.2	Gage
121	Length from rear face to rear basic diameter on boattail		0.40%	3.2	Gage
122	Length from front face to relief of ogive		0.40%	3.2	Gage
123	Length from rear face to bottom of cavity		0.40%	3.2	Gage
124	Length from front face to rear of bourrelet relief		0.40%	3.2	Gage
125	Length from front face to forward face of bourrelet relief		0.40%	3.2	Gage
126	Thickness of body wall in ogive area, min. (2 locations)		0.40%	3.2	Gage
127	Variation in wall thickness in ogive area		0.40%	3.2	Gage
128	Variation in wall thickness in bourrelet relief		0.40%	3.2	Gage (Note a)
129	Pitch diameter of thread, max.		0.40%	3.2	Gage
130	Minor diameter of thread, max.		0.40%	3.2	Gage
131	Length of effective thread, min.		0.40%	3.2	Gage

NOTES:

a. Wall variation will be checked in four locations of bourrelet relief, approximately 25 millimeters between measurements.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
4.4.2.6	Body, Machined				9326053
					NEXT HIGHER ASSEMBLY
					9326054
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
132	Foreign matter in cavity		0.40%	3.2	Gage
133	Length total		0.40%	3.16.6	Visual
134	Runout of boattail cavity		0.40%	3.2	Gage
135	Runout aft of band seat		0.40%	3.2	Gage
136	Diameter aft of band seat		0.40%	3.2	Gage
MINOR					
201	Diameter of thread chamfer		0.65%	3.2	Gage
202	Diameter of rear of boattail (I.D.)		0.65%	3.2	Gage
203	Evidence of shot or sandblasting on rotating band seat		0.65%	3.8.1	Visual
204	Evidence of poor workmanship		0.65%	3.16	Visual
205	Radii of chamfers missing or incorrect		0.65%	3.2	Visual
206	Pitch diameter of thread, min.		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Body Assembly	1 OF 2	9326054
			NEXT HIGHER ASSEMBLY 9326055

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Metal Defective		100%	3.9	Visual
2	Magnetic Particle	See 4.4.3.7		3.2	4.5.10
3	Magnetic Particle Stamp missing		100%	3.2	Visual
MAJOR					
101	Diameter of rotating band ribs		0.40%	3.2	Gage
102	Runout of rotating band with bourrelet		0.40%	3.2	Gage
103	Diameter obturator band seat		0.40%	3.2	Gage
104	Width obturator band seat		0.40%	3.2	Gage
105	Location of cannelure (3 locations)		0.40%	3.2	Gage
106	Diameter of rotating band cannelure (3 locations)		0.40%	3.2	Gage
107	Location of rotating band		0.40%	3.2	Gage
108	Band seating tightness		100%	3.8.5	4.5.5.1
109	Rotating band mechanical properties	(Note a)		3.7.2	4.5.5.2
110	Rotating band sidewall contact	(Note a)		3.8	4.5.5.2
111	Weight		0.40%	3.2	Gage
112	Foreign matter in band seat		0.40%	3.8.1	Visual
113	Location of basic diameter (Rotating Band)		0.40%	3.2	Gage

NOTES:

a. See Paragraph 4.4.3.2.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.7	Body Assembly				9326054
					NEXT HIGHER ASSEMBLY 9326055
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Width of undercuts forward and rear of rotating band (if applicable)		0.65%	3.2	Gage
202	Depth of undercuts forward and rear of rotating band (if applicable)		0.65%	3.2	Gage
203	Length, total, max.		0.65%	3.2	Gage
204	Stamping missing, misleading, or unidentifiable		0.65%	3.2	Visual
205	Location of base cover		0.65%	3.2	Gage
206	Radii or chamfers missing or incorrect		0.65%	3.2	Visual
207	Soundness of base cover weld	(Note a)		3.16	4.5.6
208	Evidence of poor workmanship		0.65%	3.16	Visual

NOTES:

a. See Paragraph 4.4.3.3.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.8	Projectile, HE, 155MM, XM795 Metal Parts Assembly		9326055
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL					
1	Pool of paint in cavity		100%	3.11.2	Visual (Probe Light)
2	Metal Defective		100%	3.9	Visual (Probe Light)
3	Proving Ground (Metal Parts Security)	(Note a)		3.6	4.5.9
MAJOR					
101	Diameter of bourrelet, max.		100%	3.2	Gage
102	Rotating band damaged		0.40%	3.16.7	Visual
103	Foreign matter in cavity		0.40%	3.11.1	Visual
104	Salt Spray Test	(Note b)		3.2	4.5.8
MINOR					
201	Paint on threads		0.65%	3.2	Visual
202	Paint on rotating band bearing surface		0.65%	3.2	Visual
203	Protective coating in cavity inadequate or damaged		0.65%	3.11	Visual
204	Protective coating on exterior surfaces inadequate or damaged	(Note c)		3.2	4.5.7
205	Corrosivee preventative compound missing from threads		0.65%	3.2	Visual
206	Marking missing, incorrect or illegible		0.65%	3.2	Visual

NOTES:

a. See Paragraph 4.4.3.6
b. See Paragraph 4.4.3.5
c. See Paragraph 4.4.3.4

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.8	Projectile, HE, 155MM, XM795 Metal Parts Assembly				9326055
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	Grommet missing		0.65%	3.2	Visual
208	Lifting plug missing or improperly seated		0.65%	3.2	Visual
209	Evidence of poor workmanship		0.65%	3.16	Visual
210	Obturator band missing		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

DOD-P-63245A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Packing and Marking for Pallet Wood Ammunition for Projectile, 155MM, HE, XM795	1 OF 1	9329574
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None Defined.				
MAJOR	None Defined.				
MINOR					
201	Pallet damaged		0.65%	3.2	Visual
202	Pallet improperly packed		0.65%	3.2	Visual
203	Strapping missing, loose or broken		0.65%	3.2	Visual
204	Marking missing, incorrect or illegible		0.65%	3.16	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Mechanical properties of body

4.4.3.1.1 Hardness. In order to determine uniformity of heat treatment, each body in the heat treatment lot (as defined by 4.4.1) shall be subjected to a hardness test approved by the procuring activity.

4.4.3.1.2 Tension test. Two projectile bodies, one of the hardest and one of the softest as determined by the hardness test shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the supplier, a production control sample may be utilized to establish a hardness range. When a production control sample is used, the hardness range thus established shall be maintained throughout the heat treatment lot. The size of the sample shall be determined by the contractor. Projectile bodies which fall outside of the established range may be re-heat treated to fall within the range or new mechanical properties may be determined from newly selected projectile bodies which will represent the true extremes of hardness in the heat treatment lot. Regardless of whether a hardness range is used or not, the high and low hardness projectiles from each heat treatment lot shall be subjected to tension testing. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the heat treatment lot represented.

4.4.3.1.3 Slow bend precrack charpy test. The two projectile bodies used for the tension test shall be used for informational purposes.

4.4.3.2 Rotating band mechanical properties and band seating sidewall contact. One body assembly shall be selected from each 8-hour production or fraction thereof from each banding machine in operation to determine compliance with applicable requirements. Failure to comply shall be cause for rejection of the assemblies represented by the sample.

4.4.3.3 Base cover weld. Five body assemblies from each one hours' production or fraction thereof per welding machine in operation shall be subjected to this test. Failure of the weld to comply with the applicable requirements shall be cause for rejection of the production represented by the sample.

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4.4.3.4 Cleaning and application of protective coating. The sample size and acceptance criteria for the inspection of cleaning the item and the application of protective coating shall be as specified in the applicable specification. Failure to comply with the applicable requirements shall be cause for taking the action recommended therein.

4.4.3.5 Salt spray. The sample size and acceptance criteria for salt spray testing shall be in accordance with the applicable specification. Failure to comply with the requirements of the applicable drawing shall be cause for taking the action recommended by the applicable drawing.

4.4.3.6 Proving ground. Security.

4.4.3.6.1 Initial production lots. Beginning with the first lot produced and continuing until three consecutive lots have complied with all of the applicable requirements specified, twenty (20) sample projectiles shall be randomly selected for this test. The twenty (20) sample projectiles shall be observed for security of parts. If any of the sample projectiles fail to comply with the applicable requirements, the lot shall be rejected.

4.4.3.6.2 Subsequent production lots. After three consecutive lots have complied with 4.4.3.6.1, ten (10) sample projectiles shall be randomly selected from each lot for this test. The sample projectiles shall be observed for security of parts. If any of the sample projectiles fail to comply with the requirements for security of parts, the lot shall be rejected and sampling will return to 4.4.3.6.1.

4.4.3.7 Magnetic particle inspection. Heat treated bodies after final machining and prior to further processing shall be inspected 100% by the magnetic particle inspection technique approved by the responsible Technical Agency. Should a controversy arise as to the validity of an indication, the manufacturer shall use metallographic test methods or an approved nondestructive method for proof of acceptability to the responsible Technical Agency.

4.4.3.8 Mult temperature. One mult shall be checked for maximum temperature during each hour, or fraction thereof, of furnace operation. The inspection point and method shall require approval of the Government technical agency. If any mult checked exceeds the applicable requirement, all mults in that heat treatment batch shall be rejected.

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4.4.4 Inspection equipment. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and paragraph 6.3 herein.

4.5 Test methods and procedures.

4.5.1 Hydrostatic test. The unpainted projectile shall be subjected to the pressure specified on the applicable drawing. A hydrostatic test plug shall be designed in accordance with 4.4.4. This plug shall be designed to engage on the nose thread and to seat on the front face of the projectile. A thread relief of 6.35mm plus or minus 2.54mm shall be used to assure that the forward threads of the projectile are not engaged when the plug is properly seated. Any type hydraulic coupling may be used provided the minimum cross-sectional area of the hydraulic feed and return in the test plug is less than 38.8 sq. mm. In addition the test equipment shall have a piezo electric gage to measure pressure-time traces for all projectiles and an adequate record shall be maintained of all results. Any projectile which fails to comply with the applicable requirements shall be classed defective and removed from the lot.

4.5.2 Metal defects. The internal cavity of the projectile shall be inspected visually prior to nosing using peripheral lights to assure conformance with the applicable requirements. Visual standards, in accordance with applicable requirements shall be set up for use during this inspection. Any projectile failing to meet the applicable requirements shall be classed defective and removed from the lot.

4.5.3 Ultrasonics

4.5.3.1 Inspection. The projectile shall be ultrasonically inspected prior to cutting band seat and nose threads. The projectile shall have complete metal parts coverage. Any projectile failing to meet the applicable requirements shall be classed defective and removed from the lot.

4.5.3.2 Equipment and procedures. The equipment and procedures for ultrasonic inspection shall be in accordance with drawing 9280269.

4.5.3.3 Ultrasonic standards. Ultrasonic inspection standards shall be provided by the contractor for use during the ultrasonic inspection. These standards shall be machined in accordance with drawings 9280308 and 9280309 and shall be inspected for all drawing requirements to determine their

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acceptability for use. Government verification of all inspections performed on the standards is required. In addition, a certificate of conformance for the standards shall be submitted to: Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801, prior to use of the standards.

4.5.3.4 Operator qualification. Personnel performing the ultrasonic inspection shall be certified in accordance with MIL-STD-410 with the following exception:

Each facility shall have as a minimum an individual who qualifies as a Level II inspector in all respects but need not include experience.

4.5.4 Mechanical properties of body

4.5.4.1 Hardness. After final heat treatment, but prior to final machining, a Brinell-type hardness test shall be performed on the outside surfaces at the band seat location.

4.5.4.2 Tension test. From the samples provided by the heat treat lots, two tension specimens conforming to the largest obtainable Standard round type specimens shall be tested in accordance with the procedure prescribed by ASTM Method E-8. The two specimens shall be treated as follows:

Base Section - The base section tensile specimen shall be taken transversely, 2 centimeters from the center line. Failure to meet the yield strength or elongation requirements of Dwg. 9326053 shall require a retest from the remaining portion of the base section. Results of the retest specimen and the first failing specimen shall be averaged to determine compliance with the applicable drawing requirement. Failure of the average results to meet the drawing requirement shall be cause for rejection of the heat-treated inspection lot.

Side Wall - Two longitudinal specimens shall be taken from the sidewall approximately midway between the nose and the base of the projectile. Failuare to meet the yield strength or elongation requirements of Dwg. 9326053 shall be cause for rejection of the heat-treated inspection lot.

Nose Section - Two longitudinal specimens shall be taken from the ogive, not more than 73 millimeters from front face. Failure to meet the yield strength or elongation requirement of Dwg. 9326053 shall be cause for rejection of the heat treated inspection lot.

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4.5.4.3 Slow bend precrack charpy test. The test shall be performed at two temperatures; one at room temperature and the other at -54°C (Celcius). The body wall samples shall be taken circumferencially from the side wall of the body at midwall. Notches cut into the sample shall be so oriented that they run parallel to the projectile longitudinal axis. The base specimen samples will be taken transversely from the base and notches cut into these samples will be oriented perpendicular to the longitudinal axis of the projectile. Testing shall be in accordance with the test procedure defined in the Appendix of this specification. The contractor shall submit all test reports monthly in accordance with Data Item Description DI-T-1909, reporting all information required by the Appendix. For any further information pertaining to this test, contact: Commander, ARRADCOM, ATTN: DRDAR-SCM-E, Dover, New Jersey 07801.

4.5.5 Rotating band.

4.5.5.1 Band seating tightness. Each sample shall be tested utilizing the equipment and procedures prescribed in Frankford Arsenal Detroit Testing Machine Manual (see 2.1). Indentor type and ram pressure to be as specified for "155MM, M107". Indentor location to be between forward cannelures.

4.5.5.2 Mechanical properties and sidewall contact. The samples shall be sectioned by cutting through two planes normal to the longitudinal axis one centimeter forward and rear of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for inspection for mechanical properties and the other section for inspection of sidewall contact.

4.5.5.2.1 Mechanical properties. The section provided shall be flattened without hammering so as to hold to a minimum any additional cold working. The specimen blank shall be machined so that all serration marks from the band seat are removed. A test specimen having two centimeter gage length and a one centimeter gage width, and grip ends of sufficient length to accomodate the tensile machine, shall be removed from the blank band. The specimen shall be tested in accordance with the procedure prescribed by ASTM Method E-8.

4.5.5.2.2 Sidewall contact. The specimen provided shall be sectioned into two parts exposing four surfaces, each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove metal smear effects of saw cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements.

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4.5.6 Base cover weld. The test of the base cover weld shall be performed as follows: Visually inspect the base cover and select the area that appears to have the weakest weld; insert a chisel in this area and apply force by use of a one pound hammer and an approximate one foot stroke. If the weld breaks loose after 2 strokes, the body assembly will be classed defective. If the weld is found to be secure, the burr made by the chisel may be knocked back into position and accepted.

4.5.7 Protective coating. The procedure for testing the cleaning and surface treatment methods for components and assemblies shall be in accordance with the methods and criteria for testing as specified in the applicable specification.

4.5.8 Salt spray. The salt spray test prescribed by the applicable drawing shall be performed utilizing the equipment and procedures specified by ASTM Method B-117 and TT-C-490.

4.5.9 Proving ground. These tests shall be performed at a Government proving ground in accordance with the applicable acceptance test procedure. The sample projectiles selected shall be inert-loaded within 50 grams (0.1 pound) of an as-fired weight (with fuze) of 46.90 kilograms (103.3 pounds) and fired in a M185 or M199 series cannon using a PXR-6297 proof charge (see 6.11) which produces an average piezo equivalent pressure of 407.0 MPa (59,000 psi). Chamber pressures will be based on copper pressure gages. The projectiles and propelling charges shall be uniformly temperature conditioned to 63° Celsius (145°F) minus 3° Celsius (5°F) for a minimum of 24 hours prior to testing. Pressure-time traces and differential pressures shall be obtained for information only by means of a spindle tap in the breech of the cannon and a piezo gage at the projectile base, location for each projectile fired. Adequate photographic and instrumental coverage shall be provided to assure compliance with requirements. Range and deflections shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts for informational purposes.

4.5.10 Magnetic particle inspection. The procedures for performing this test utilizing the wet flourescent method shall be as specified in Specification MIL-I-6868, for indirect magnetization by either the continuous method or the residual method. The suspension shall be applied to the entire body, inside and outside. After draining, the body shall be thoroughly inspected in a darkened area under the ultra-violet light which will produce adequate flourescence of the magnetic particles. The solids content of the suspension shall be

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maintained at a level recommended by the manufacturer. The suspension shall be checked at the start of each days' production and at intervals of not over 4 hours continuous operation, utilizing the procedures and the equipment prescribed by Specification MIL-I-6868. When performing the test for solids content of suspension, if it is determined that it does not comply, all assemblies that have been accepted since the last satisfactory test shall be retested after the suspension has been brought to the proper level. After acceptance, the body assembly shall be demagnetized and cleaned. The magnetic particle test shall be performed on equipment that complies with the requirements of Specification MIL-I-6867. The operator(s) of the equipment and the inspectors making the determination shall be certified in accordance with MIL-STD-410.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. The projectile shall be packed in accordance with Dwg. 9329574.

5.1.1 Level A & B. Level A & B apply.

5.1.2 Level B-1. Projectile metal parts assemblies shall be palletized in accordance with Dwg. 9329574.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 155MM, HE, XM795 Projectile.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection procedures and/or equipment designs for approval - See MIL-A-48078. Submit equipment designs as required to: Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.4 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

6.5 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to: Commander, ARRADCOM, ATTN: DRDAR-QAR-Q and DRDAR-LCU-CP, Dover, New Jersey 07801.

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6.6 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible Government technical agency that their effectiveness is equal to, or better than the quality assurance provisions specified herein.

6.7 Band seating. The operating procedures for use of Detroit Testing Machine as prepared by the Inspection Equipment Branch, US Army, Frankford Arsenal, DRDAR-FA.

6.8 Corrosion preventive compounds. The following materials complying with specification MIL-C-16173D, Grade 2, have been found to be compatible with TNT.

a. Cosmoline No. 1102 - E.F. Houghton Co., Phila. PA

b. Non-Rust No. 207, Grade 2 - Daubert Chemical Co., Chicago, IL

c. Petrotest 2 - Pennsylvania Refining Co., Butler, PA

6.9 Variables. In the application of alloy steel, blooms, billets, slabs and bars, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation dimensions in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices and processing techniques having capabilities of compensating for such variables.

6.10 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.11 Proof charge pressure. Prior to proving ground acceptance testing, five (5) proof charges, PXR-6297 will be fired in the cannon to be utilized in the acceptance test. Charges will be conditioned the same as acceptance test

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charges. Charges to be utilized for the acceptance test will then be propellant weight adjusted to produce the desired pressure based on the 5 charge sample and the charge weight to pressure slope as established in the propelling charge assessment test. If the average pressure of the 5 charge sample is within 6.5 MPa (1000 psi) of the desired pressure, no weight adjust will be made.

6.12 Hydrostatic test plug. Examination should be made of a hydrostatic plug after failure of a body. A badly nicked or damaged plug may cause failure of an acceptable body.

6.13 Data requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	Data Requirement	Applicable DID
4.5.4.3	Test Report	DI-T-1906

(Copies of data item descriptions required by the contractors in conjunction with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project No.: 1320-A566

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APPENDIX

TEST METHOD FOR SLOW BEND PRECRACKED CHARPY TEST

10. SCOPE

10.1 Scope. This appendix covers the test method used in determining the nominal strength of a Charpy specimen containing a fatigue crack and tested in slow bending.

20. APPLICABLE DOCUMENTS

20.1 ASTM Standards.

E-4	-Verification of Testing Machines
E-139	-Recommend Practice for Conducting Creep-Rupture and Stress Rupture Tests of Metallic Materials
E-399	-Plane Strain Fracture Toughness of Metallic Materials

30. SUMMARY OF METHOD

30.1 The method involves three point bend testing of notches and fatigue cracked specimens using fixtures which minimize the contribution of friction forces to the measured applied load. The maximum load is recorded and the nominal crack strength is determined from this value and the original dimensions of the specimen using the simple beam equation.

40. PRECAUTIONARY NOTES

40.1 Precautionary notes. When preparing specimens and testing for nominal crack strength, it is important that certain precautions be kept in mind.

40.1.1 Crack strength. The crack strength of some materials will be sensitive to the rate of loading, and loading rates should be confined to the range given in 80.3.2.

40.1.2 Temperature. The crack strength will be influenced by the testing temperature, and calibration relations may change with testing temperature.

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50. DEFINITIONS

50.1 Nominal crack strength. C. as determined by this method, the nominal crack strength is defined as the nominal stress in a notched and cracked bend specimen calculated on the basis of the length of the initial uncracked ligament and the maximum load sustained in the test. The crack strength is analogous to the ultimate tensile strength in that it is based on the dimensions of the specimen before testing.

50.2 The following symbols are used in this test method.

a - Fatigue crack depth (In)

B - Specimen thickness (In)

W - Specimen depth (In)

M - Bending Moment (In - lb_f)

f(a/W) - Power series of ratio (a/W) from ASTM-E-399

c - Nominal Crack Strength - (KSI)

P_{max} - Maximum Load Applied - (lb_f)

S - Span between Support Rollers - (In)

$K_f(max)$ - Maximum Stress Intensity Factor (KSI In)

K - Stress Intensity Factor (KSI In)

60. APPARATUS

60.1 Tension testing machine. Tension Testing Machine conforming to the requirements of ASTM E-4.

60.2 Bend-Test fixture. The procedures involve the testing of the specimen in three point bending. The bend fixture should be so designed as to minimize errors that arise from friction between the specimen and the supports. This can be accomplished by allowing the support rollers to rotate and move apart slightly during loading as described in ASTM E-399 (Figure 1). The nominal span shall be 1.97 in. If the diameter of the support rollers is between 0.30 in. to 0.40 in., there will be a negligible change in the span during deformation of the specimen.

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60.3 Temperature control. The testing temperatures shall be controlled by use of an integral temperature conditioning chamber as part of the testing apparatus. Any suitable method may be used to temperature condition the specimen providing the region, including the notched and cracked section, can be uniformly maintained within $\pm 2^{o}F$ of the desired test temperature.

60.4 Temperature measurement. The temperature of the specimen during testing shall be measured by means of two thermocouples attached on either side of the notches and cracked section. These thermocouples may be spot welded in place, however, the spot weld locations should avoid the crack path. At the test temperature, the difference between the temperatures indicated by these two thermocouples should not exceed $1^{o}F$ and the average of these two temperatures shall be within $\pm 2^{o}F$ of the desired test temperature.

60.4.1 Thermocouples. Temperatures shall be measured with calibrated thermocouples used in conjunction with potentiometers or millivoltmeters. Such measurements are subject to various errors; reference should be made to ASTM E-139 for a discussion of errors.

60.4.2 Calibration. The temperature measuring apparatus should be calibrated periodically against standards traceable to the National Bureau of Standards in order to ensure that accuracy specified above can be achieved.

70. TEST SPECIMENS, DIMENSIONS AND PREPARATION

70.1 Dimensions. The precracked Charpy bend specimen is shown in Figure 1. The crack depth (including notch), a, is nominally equal to one-half the thickness, B, and shall lie within the range between 0.40 to 0.55B. $\frac{}{2}$

70.1.1 Crack starter. The crack starter may consist of a "V" notch having an included angle between 45° and 60° provided with a small radius at its tip. Experience has shown that a notch radius of less than 0.003 in. is necessary in order to facilitate the initiation of a fatigue crack at a low stress intensity level.

70.2 Fatigue cracking. In order to ensure a sufficiently sharp fatigue crack and to facilitate the production of this crack, the following procedure shall be followed.

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70.2.1 Specimen condition. The fatigue cracking shall be conducted with the specimen in the condition of heat treatment in which it is to be tested.

70.2.2 Cracked method. The method used for fatigue cracking shall be such that the crack extension direction is perpendicular to the notched face of the specimen. Cantilever bending or three point bending may be used to produce fatigue cracks. Cantileer bending facilitates completely reversed loading which aids in starting and propagating the crack. For practical purposes, the K calibration for three point bending may be used to estimate the stress intensity factors.

NOTE 1: The expression given in ASTM E-399 for the K calibration in three point bending may be rewritten in terms of the bending moment M as follows and used to compute the stress intensity in cantilever bending: $K = (4M/BW^{3/2}) f(a/W)$ where the values of the function f(a/W) are given in tabular form in ASTM E-399.

NOTE 2: The stress distribution at the crack tip in cantilever bending can be influenced by excessive clamping forces to the extent that the fatigue crack will deviate from a plane normal to the notched face of the specimen.

70.2.3 Cracking. During the last 0.020 in. of fatigue cracking, the maximum intensity of the fatigue cycle, Kf (max) shall not exceed 20 $KpI-^{3/2}$. The stress intensity range should not be less than 0.9Kf (max).

80. TEST PROCEDURE

80.1 Number of tests. Samples shall be selected as specified in the basic specification.

80.2 Specimen measurement. All specimen dimensions shall be within the tolerances shown in Figure 1.

80.2.1 Measure the thickness, B, to the nearest 0.001 in. at not less than two positions between the fatigue crack tip and the unnotched edge of the specimen and record the average positions, one on each side of the crack starter notch and record the average value.

80.2.2 After fracture measure, the initial fatigue crack depth, a, to the nearest 0.5 percent at the following three positions: at the center of the crack and midway between the

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center and the inter-section of the crack front within the specimen surfaces. Use the average of these three measurements as the crack depth in the calculation of C (see 90.1). If the difference between any two of the crack length measurements exceeds 5 percent of the average, or if part of the crack front is closer to the machined notch root than 5 percent of the average, the specimen should be discarded. Also, if the length of either surface trace of the crack is less than 80 percent of the average crack length, as defined above, the specimen should be discarded.

80.3 Bend testing. Set up the bend test fixture so that the line of action of the applied load shall pass midway between the support role centers within 0.025 in. of the distance between these centers. Measure the span (distance between the support role centers) within 1 percent and record this value as S. This value should fall between 1.9 in. and 2.1 in. If not, the span should be adjusted to fall within this range.

80.3.1 Locate the specimen with the crack tip midway between the foll centers to within 0.025 in., and square to the roll axes within 2 degrees. Check the parallelism between the back face of the specimen and the loading roller by sighting the gap between these surfaces just before contact is made. As a small load is applied, light in this gap should estinguish simultaneously across the thickness of the specimen. If this condition is not met, the loading roller should be suitably adjusted.

80.3.2 Load the specimen at the rate between 700 lbf/min. and 3600 lbf/min. until the maximum load is reached and record this value as Pmax (lbf).

NOTE 3: It is desirable to follow the load change carefully because specimens having low toughness will break abruptly and no slow decrease in load beyond the maximum will be observed.

90. CALCULATION AND INTERPRETATION OF RESULTS

90.1 Calculations. Calculate the nominal crack strength of the specimen in KSI from the measured value of Pmax and the initial dimensions of the specimen as follows:

$$C = 3000\ P_{max} S/2\ B(W - a)^2$$

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90.2 Fracture appearance. The appearance of the fracture may furnish valuable supplementary information. A means for describing the fracture appearance of cracked bend speciment is given in ASTM E-399 and may be used to characterize the fracture appearance of the precracked Charpy specimens.

100. REPORT

100.1 The report shall include the following for each specimen tested:

100.1.1 All specimen dimensions measured.

100.1.2 The maximum load Pmax.

100.1.3 The nominal crack strength.

100.1.4 Test temperature

100.1.5 The maximum stress intensity $K_{f(max)}$ during the final 0.020 in. of fatigue cracking.

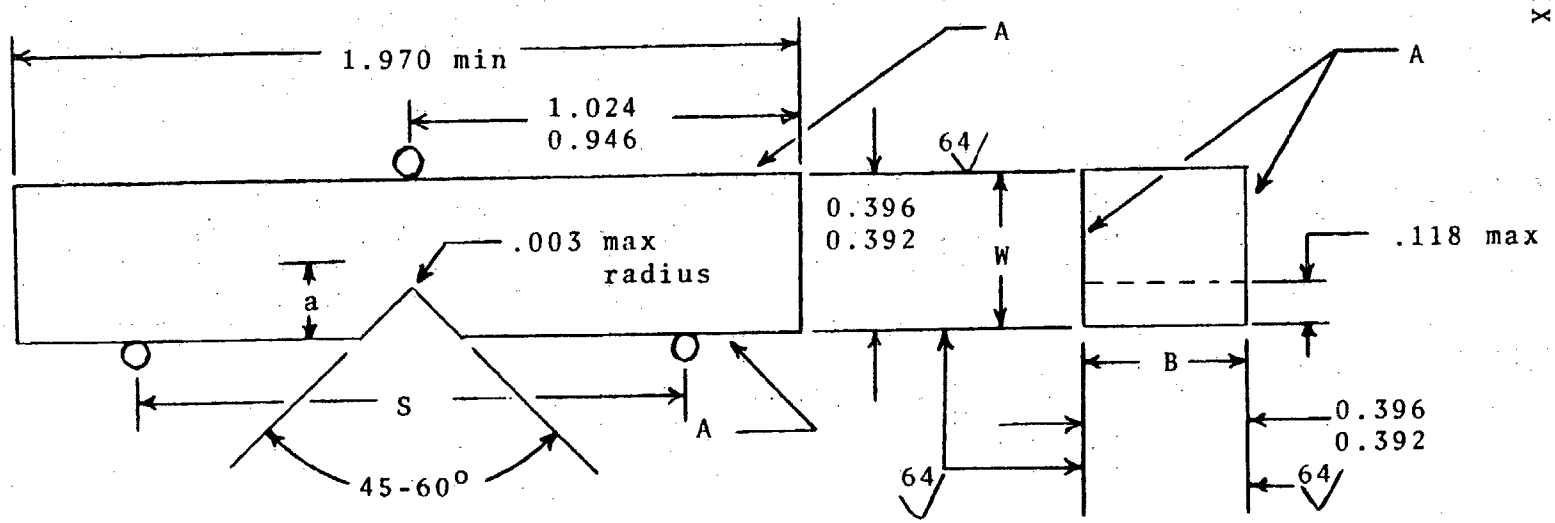
100.1.6 Crack plane orientation.

100.1.7 Fracture appearance.

100.1.8 0.2 percent offset tensile yield strength.

100.1.9 Fracture toughness ($K_Q = K_{1c}$)

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NOTE 1: All dimensions in inches.

NOTE 2: "A" surfaces shall be perpendicular or parallel as applicable to within 0.001 In TIR.

NOTE 3: Crack starter shall be perpendicular to specimen length and thickness to within ±2 degrees.

FIGURE 1. Dimensions of Precracked Charpy Specimens

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